25901098 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBIF200 - BIOSTATISTICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Select the modal value from 9, 9, 10, 11. [A] / [CO5]
2. Identify one demerit of arithmetic mean. [A] / [CO5]
3. Explain the meaning of P(A|B) in words. [K] / [CO7]
4. What is a p-value? [K] / [CO4]
5. What is the purpose of using map diagrams in data presentation? [U] / [CO1]
6. What is primary data? [U] / [CO1]
7. Plan the procedure to compute Harmonic Mean for 3, 6 and 9. [A] / [CO5]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

8. Explain the addition theorem for three events. [U] / [CO7]

9. Utilize the formula QD = (Q3 − Q1) / 2 to find quartile deviation for the data: 10, 20, 30, 40, 50, 60. [A] / [CO6]

10. Discuss the applications of biostatistics in clinical research and public health, highlighting its role in decision-making. [U] / [CO1]

11. What is a statistic in sampling? How does it relate to a sample? [U] / [CO4]

12. Classify partition values into their types and define each. [U] / [CO5]

13. Explain mutually exclusive and equally likely events with examples. [U] / [CO7]

14. Why is a larger sample size preferred when estimating a parameter? [K] / [CO4]

15. Explain the difference between addition rule and multiplication rule with examples. [U] / [CO7]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **3** questions

Each question carries **10** marks

16. Write short notes on kite diagram and dot diagram and mention where each is used. [K] / [CO2]

17. State the difference between variance and standard deviation with an example. [K] / [CO6]

18. Define probability. Explain classical probability and statistical probability with suitable examples. [U] / [CO7]

19. Explain the applications of t-test in biological experiments. [U] / [CO4]

20. Describe the step-by-step procedure for constructing class intervals from a raw dataset. Discuss the factors to be considered while deciding the number and width of class intervals. [U] / [CO1]

[10x3 = 30]